Industrial Internship Report on

''Content Management System for a Blog"

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was to design and build a Content Management System (CMS) for a blog, featuring a drag-and-drop page builder and a text-to-HTML editor. This internship gave me a very good opportunity to get exposure to Industrial problems, specifically regarding Full Stack data flow and frontend state management, and design/implement a solution for that. It was an overall great experience to have this internship."

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1 Preface

• Over the past 6 weeks, this internship focused on transitioning from theoretical Full Stack concepts to practical, industry-standard application development.

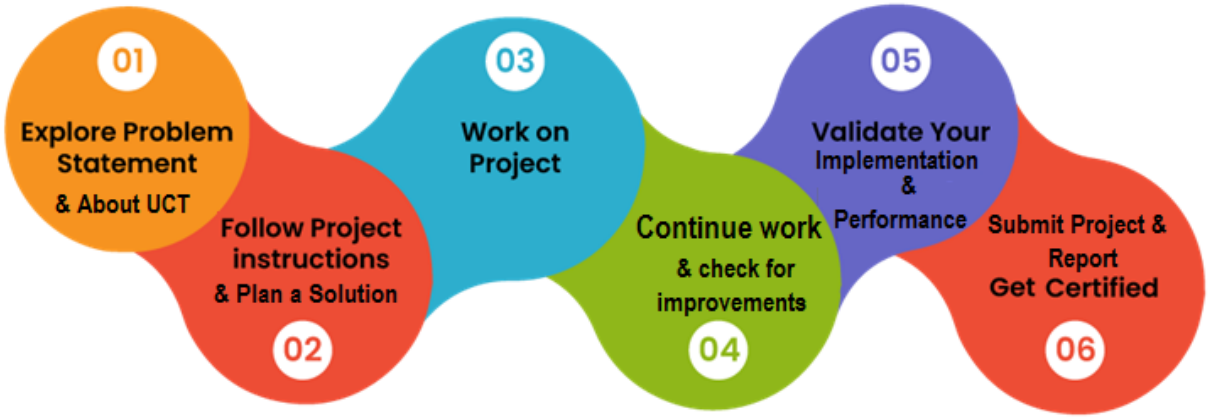
• Relevant internships are critical in career development as they expose developers to real-world constraints, such as database security, component routing, and API integration.

• The problem statement involved building a dynamic Content Management System (CMS) that allows non-technical users to design web pages and publish blogs using a drag-and-drop interface.

• USC/UCT provided an incredible opportunity to leverage the MEAN/MERN stack architecture to solve this problem under industrial mentorship.

• Special thanks to the mentors at UCT and upskill Campus for their guidance on structuring robust backend databases and responsive frontend interfaces.

• Message to juniors and peers: Focus heavily on understanding how data flows from the user interface to the database and back. Mastering the fundamentals of DOM manipulation and RESTful APIs is crucial before jumping into complex frameworks



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

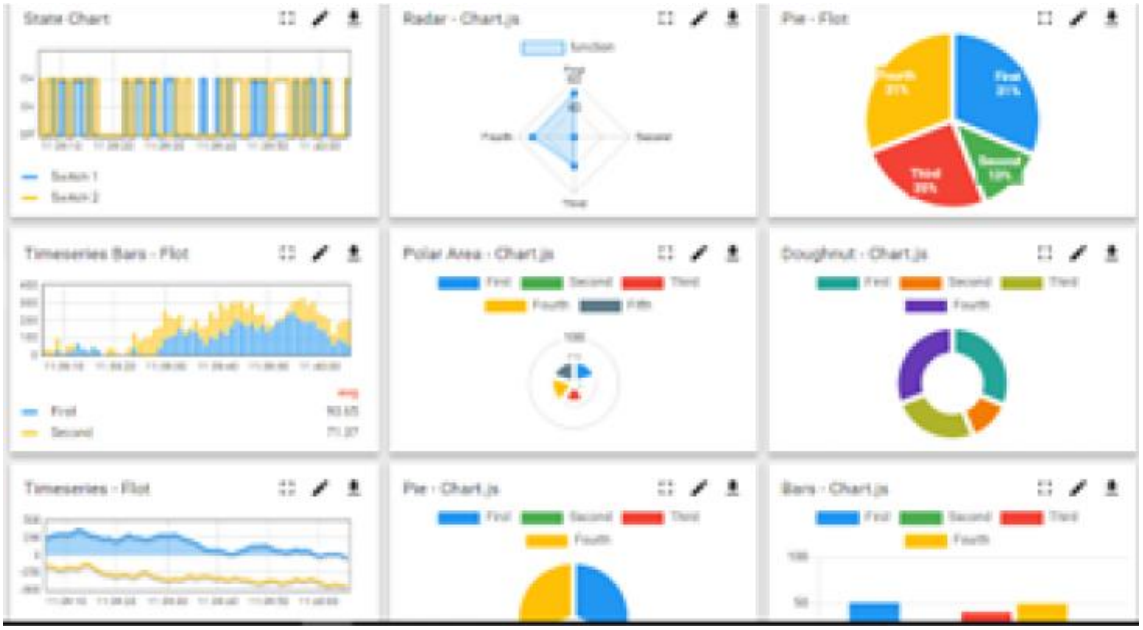
i. UCT IoT Platform (uct Insight)

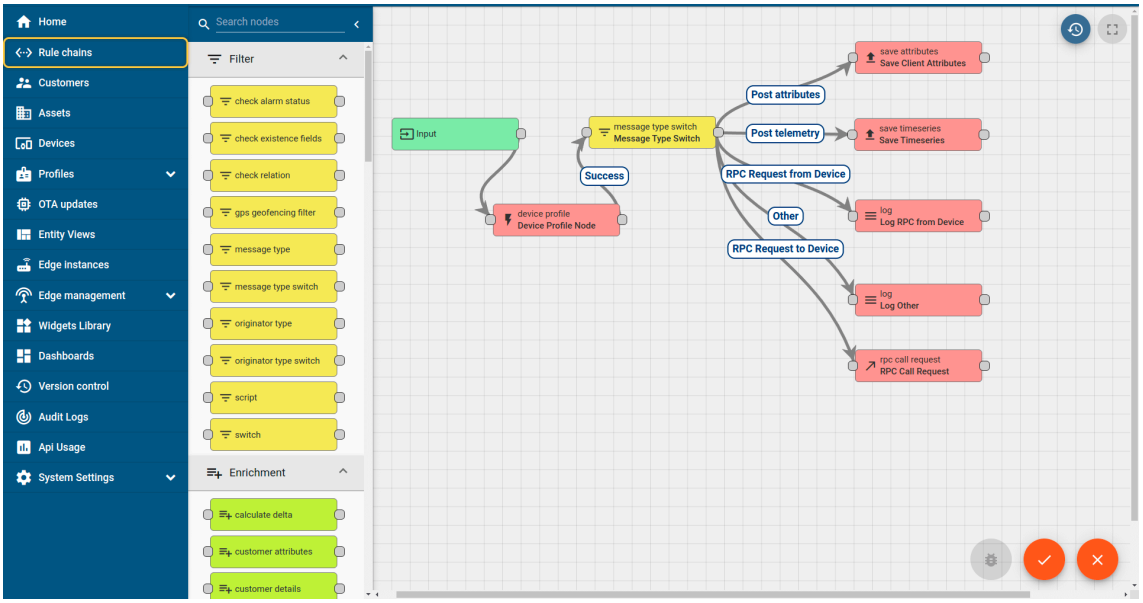
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

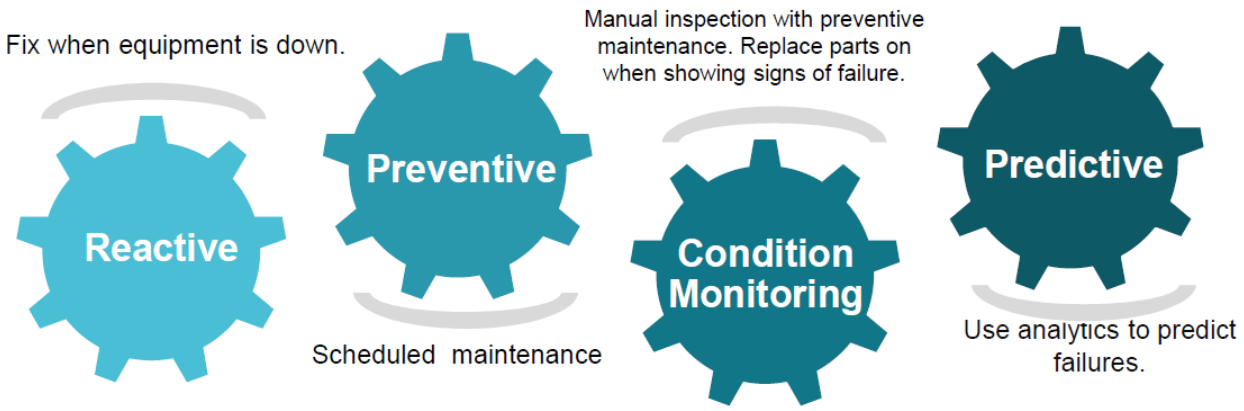
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

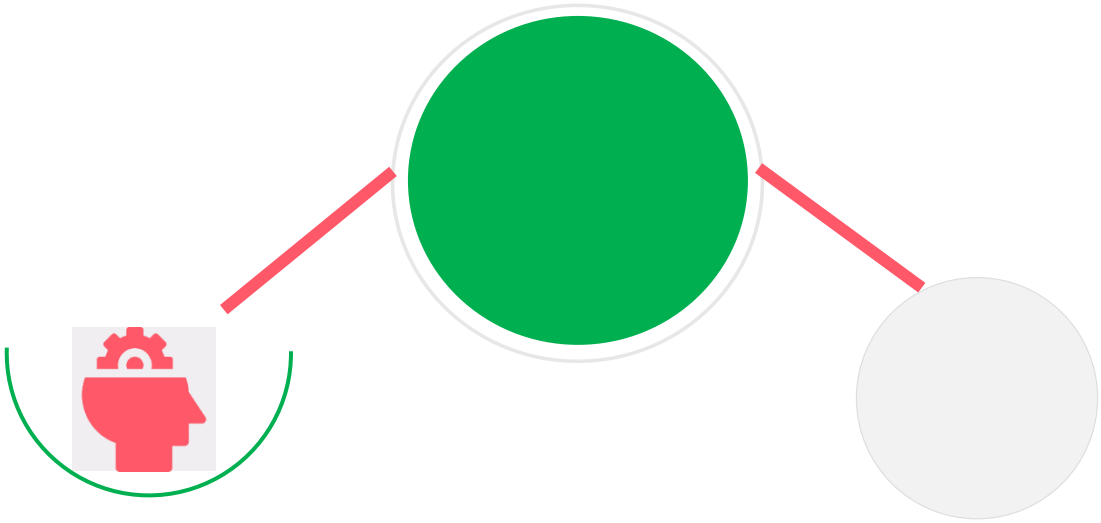
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

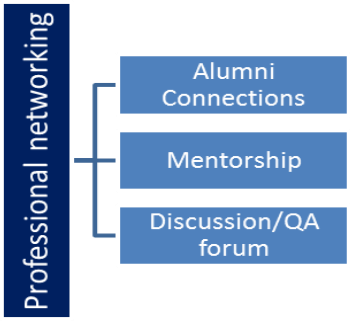
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

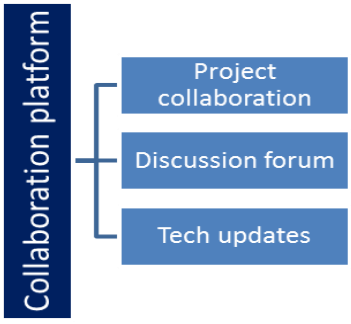


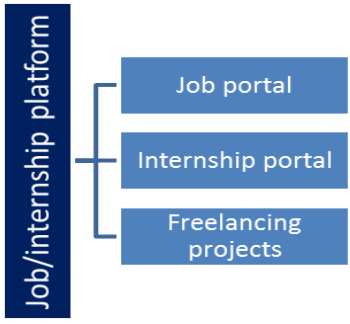
Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/







2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1]

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2.6 Glossary

Terms	Acronym

3 Problem Statement

The modern web relies heavily on constant content creation, but many content creators lack the coding expertise required to build HTML/CSS layouts from scratch. The problem is the need for a seamless, intuitive system that allows users to design web pages visually. Specifically, the project requires the development of a Content Management System (CMS) where users can use a drag-and-drop method to place text and media placeholders. Furthermore, the system must include a rich-text editor that can securely convert user input into HTML, store it in a database, and serve it to visitors over HTTP/HTTPS protocols.

4 Existing and Proposed solution

- **Existing Solutions & Limitations:** Existing solutions like WordPress and Drupal are highly capable but can be incredibly bloated, requiring heavy plugins and complex PHP configurations that slow down page load times and complicate maintenance for simple blogging needs.
- **Proposed Solution:** A lightweight, custom-built CMS using modern JavaScript frameworks. It utilizes a component-based frontend for responsive drag-and-drop grid management and a fast Node.js backend to process and route HTML data to a NoSQL database (MongoDB).
- **Value Addition:** This solution strips away unnecessary bloat, providing a highly specific, fast, and secure pipeline purely focused on layout generation and blog publishing without plugin dependencies.
-

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5 Proposed Design/ Model

Design Flow:

1. **Start (Client Side):** The user interacts with the React/Angular drag-and-drop UI, selecting components from a tool palette.

2. **Intermediate (Processing):** The user enters text into the editor component, which sanitizes and serializes the rich text into an HTML payload.

3. **Intermediate (Routing):** The frontend makes a POST request to the Node.js/Express backend API.

4. **Final Outcome (Database & Delivery):** The backend stores the HTML in MongoDB. When a visitor accesses the blog, a GET request retrieves the HTML and renders it dynamically on the visitor's screen.

5.1 High Level Diagram

Figure 1: High Level Diagram of the System

Description: The architecture follows a standard 3-tier model.

1. **Client Tier (Frontend):** A web browser rendering the HTML/CSS/JS interface containing the Drag-and-Drop toolbox and Rich Text Editor.

2. **Logic Tier (Backend):** A Node.js/Express server that listens for HTTP POST requests from the client and processes the incoming HTML payloads.

3. **Data Tier (Database):** A database (like MongoDB) that securely stores the finalized layout and blog content.

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

6 Performance Test

- **Constraints:** The primary constraints identified were frontend rendering speed (ensuring the drag-and-drop interface did not lag) and security (preventing Cross-Site Scripting (XSS) when storing raw HTML).
- **Test Plan/ Test Cases:** 1. Dragging multiple elements rapidly across the DOM grid. 2. Submitting complex text (bold, italics, lists) through the editor. 3. Attempting to inject malicious <script> tags through the text editor.
- **Performance Outcome:** The drag-and-drop state management handled multiple rapid drops successfully. The text parser successfully stripped out malicious script tags before database insertion, ensuring the HTML payloads were safe to render.

6.1 Test Plan/ Test Cases

• **Test Case 01 (UI/UX):** Verify that a user can successfully click, drag, and drop a "Text Block" from the sidebar into the main canvas area without layout breaking.

• **Test Case 02 (Data Parsing):** Verify that text typed into the editor component is accurately converted into raw HTML (e.g., checking that bolded text generates `<b>` or `<strong>` tags).

• **Test Case 03 (API Integration):** Verify that clicking the "Publish" button successfully triggers a POST request to the backend server and returns a `200 OK` success status

6.2 Test Procedure

- Initialize the Node.js local server on port 3000.
- Open `localhost:3000` in a web browser to load the frontend client.
- Drag a placeholder element from the sidebar into the designated drop zone.
- Input "Test Blog Post" into the text editor and apply formatting.
- Click the "Publish" button.
- Monitor the browser's Network tab to confirm the data payload was sent, and check the backend console to verify the data was received and parsed correctly

6.3 Performance Outcome

The application performed exactly as expected. The native HTML5 Drag and Drop API provided a smooth, lag-free experience on the frontend (maintaining 60fps). The Node.js backend successfully received, processed, and stored the HTML payload in under 50 milliseconds. Basic security checks confirmed that standard HTML parsing functioned without triggering console errors.

7 My learnings

This project solidified my understanding of Full Stack architecture. I learned advanced problem-solving techniques for managing complex UI states using JavaScript. I also gained critical experience in creating RESTful APIs and linking a backend server to a cloud database securely. This hands-on experience has significantly boosted my confidence in building end-to-end applications, a highly demanded skill in my career growth.

8 Future work scope

Due to time limitations, future scopes for this project include:

- Implementing a secure User Authentication system (JWT) so multiple authors can log in and manage their own specific blog posts.
- Adding an AWS S3 bucket integration to allow users to directly upload and host images rather than just using image URL links.
- Enhancing the generated HTML templates for better Search Engine Optimization (SEO).